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► KerasTuner

KerasTuner

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KerasTuner is an easy-to-use, scalable hyperparameter optimization framework that solves the pain points of hyperparameter search. Easily configure your search space with a define-by-run syntax, then leverage one of the available search algorithms to find the best hyperparameter values for your models. KerasTuner comes with Bayesian Optimization, Hyperband, and Random Search algorithms built-in, and is also designed to be easy for researchers to extend in order to experiment with new search algorithms.

Quick links

- [Getting started with KerasTuner](#)
- [Developer guides](#)
- [API documentation](#)
- [KerasTuner on GitHub](#)

Installation

Install the latest release:

```
pip install keras-tuner --upgrade
```

You can also check out other versions in our [GitHub repository](#).

Quick introduction

Import KerasTuner and TensorFlow:

```
import keras_tuner
import keras
```

Write a function that creates and returns a Keras model. Use the `hp` argument to define the hyperparameters during model creation.

```
def build_model(hp):
    model = keras.Sequential()
    model.add(keras.layers.Dense(
        hp.Choice('units', [8, 16, 32]),
        activation='relu'))
    model.add(keras.layers.Dense(1, activation='relu'))
    model.compile(loss='mse')
    return model
```

Initialize a tuner (here, `RandomSearch`). We use `objective` to specify the objective to select the best models, and we use `max_trials` to specify the number of different models to try.

```
tuner = keras_tuner.RandomSearch(
    build_model,
    objective='val_loss',
    max_trials=5)
```

Start the search and get the best model:

```
tuner.search(x_train, y_train, epochs=5, validation_data=(x_val, y_val))  
best_model = tuner.get_best_models()[0]
```

To learn more about KerasTuner, check out the [getting started guide](#).

Citing KerasTuner

If KerasTuner helps your research, we appreciate your citations. Here is the BibTeX entry:

```
@misc{omalley2019kerastuner,  
  title = {KerasTuner},  
  author = {O'Malley, Tom and Bursztein, Elie and Long, James and Chollet, Fran{\c{c}ois  
and Jin, Haifeng and Invernizzi, Luca and others},  
  year = {2019},  
  howpublished = {\url{https://github.com/keras-team/keras-tuner}}  
}
```

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